`GeoModels` Tutorial: simulation, estimation and prediction of asymmetric spatial data using skew-Gaussian random fields

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Introduction

In this tutorial we show how to analyze geo-referenced spatial data using skew-Gaussian random fields (RFs), as depicted in Zhang and El-Shaarawi (2010) and Alegría et al. (2017), with the R package `GeoModels`. We first load the R libraries needed for the analysis and set the name of the model in the `GeoModels` package:

```
rm(list=ls())
require(GeoModels)
require(fields)
require(sn)
model="SkewGaussian" # model name in the GeoModels package
set.seed(8)
```

Simulation of a skew Gaussian random field

Let us consider a spatial Gaussian RF $Z = \{Z(\boldsymbol{s}), \boldsymbol{s} \in S\}$, where $\boldsymbol{s}$, in this tutorial, represents a location in a spatial domain $S \subset \mathbb{R}^2$. We assume that Z is stationary with zero mean, unit variance and correlation function given by $\rho(\boldsymbol{h}) = \mathrm{cor}(Z(\boldsymbol{s}+\boldsymbol{h}), Z(\boldsymbol{s}))$.

Then we consider the RF $Y = \{Y(\boldsymbol{s}), \boldsymbol{s} \in S\}$ defined as

$$Y(\boldsymbol{s}) = \mu(\boldsymbol{s}) + \sigma\left(\frac{\eta}{\sigma}|Z_1(\boldsymbol{s})| + Z_2(\boldsymbol{s})\right), \tag{1}$$

where Z_1 and Z_2 are independent copies of Z, $\eta \in \mathbb{R}$ is an asymmetry parameter and $\sigma > 0$ is a scale parameter. Moreover $\mu(\boldsymbol{s}) = X(\boldsymbol{s})^T\boldsymbol{\beta}$, where $X(\boldsymbol{s})$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_k)^T$ is a vector of unknown parameters (in this tutorial we fix $k = 2$).

Marginally, $Y(\boldsymbol{s})$ follows a skew-Gaussian distribution $SN(\mu(\boldsymbol{s}), \omega^2, \alpha)$ (Azzalini and Capitanio, 2014) with $\omega^2 = \eta^2 + \sigma^2$ and $\alpha = \eta/\sigma$, and density

$$f_Y(y) = \frac{2}{\omega}\phi\left(\frac{y-\mu(\boldsymbol{s})}{\omega}\right)\Phi\left(\alpha\frac{y-\mu(\boldsymbol{s})}{\omega}\right),$$

where ϕ and Φ denote the standard Gaussian density and distribution functions, respectively. Under this setting

$$\mathrm{E}(Y(\boldsymbol{s})) = \mu(\boldsymbol{s}) + \eta\sqrt{\frac{2}{\pi}}, \qquad \mathrm{var}(Y(\boldsymbol{s})) = \sigma^2 + \eta^2\left(1 - \frac{2}{\pi}\right),$$

and the correlation function is given by

$$\rho_Y(\boldsymbol{h},\eta,\sigma)=\frac{2\eta^2}{\pi\sigma^2+\eta^2(\pi-2)}\left[\sqrt{1-\rho^2(\boldsymbol{h})}+\rho(\boldsymbol{h})\arcsin\rho(\boldsymbol{h})-1\right]+\frac{\sigma^2\rho(\boldsymbol{h})}{\sigma^2+\eta^2(1-2/\pi)}. \tag{2}$$

Suppose we want to simulate a realization of Y at $N=1200$ spatial locations uniformly distributed in the unit square. We first set the spatial coordinates with associated covariate matrix:

```
N=1200 # number of locations sites
# Define the spatial coordinates
x=runif(N)
y=runif(N)
coords=cbind(x,y)
X=cbind(rep(1,N),runif(N))
```

We then specify the regression location, scale, asymmetry and nugget parameters. The names of the parameters for the skew-Gaussian RF can be obtained with the function `NuisParam`:

```
NuisParam(model,num_betas=2)
# [1] "mean" "mean1" "nugget" "sill" "skew"
```

Note that the option `num_betas` specifies the number of regression coefficients involved. We then fix the parameters:

```
mean=0.5; mean1=-0.5
sill=1.5
nugget=0
skew=-3
```

where `mean`, `mean1`, `sill`, `skew` correspond to β_1, β_2, σ^2 and η, respectively.

For the correlation function we assume the isotropic generalized Wendland class

$$\rho(\boldsymbol{h};\alpha,\delta,\kappa)=\begin{cases}\dfrac{1}{B(2\kappa,\delta+1)}\displaystyle\int_{\|\boldsymbol{h}\|/\alpha}^{1}u\left(u^2-(\|\boldsymbol{h}\|/\alpha)^2\right)^{\kappa-1}(1-u)^{\delta}\,du, & 0\le\|\boldsymbol{h}\|<\alpha,\\ 0, & \|\boldsymbol{h}\|\ge\alpha.\end{cases} \tag{3}$$

In the spatial case $d=2$, the model is valid if $\delta\ge 1.5+\kappa$ and $\kappa\ge 0$. Using asymptotic arguments, Bevilacqua et al. (2019) show that this correlation model exhibits the same

features as the Matérn correlation model. Additionally, it is compactly supported, which is an interesting feature from a computational point of view.

The names of the parameters associated with a correlation model can be obtained with the `CorrParam` function:

```
corrmodel="GenWend" ## correlation model
CorrParam(corrmodel)
# [1] "power2" "scale" "smooth"
```

Then we set the correlation parameters of the generalized Wendland model:

```
scale=0.2
smooth=1
power2=5
```

where `scale`, `smooth` and `power2` correspond to α (the compact support), κ (the smoothness parameter) and δ, respectively. In particular, $\delta = 5 \geq 1.5 + \kappa = 2.5$. Under this setting the RF is once mean-square differentiable.

We are now ready to simulate the skew-Gaussian RF using `GeoSim`:

```
param=list(mean=mean,mean1=mean1,sill=sill,
           nugget=nugget,scale=scale,skew=skew,
           power2=power2,smooth=smooth)
data=GeoSim(coordx=coords,corrmodel=corrmodel,
                      model=model,X=X,param=param)$data
```

The use of the generalized Wendland model allows the associated covariance matrix to be sparse. The covariance matrix of the skew-Gaussian random field can be computed as

```
cc=GeoCovmatrix(coordx=coords,corrmodel=corrmodel,
                  model=model,sparse=TRUE,X=X,param=param)
cc$nozero
```

The option `sparse=TRUE` permits sparse-matrix algorithms for the covariance matrix (see the `spam` package; Gerber et al., 2017). The original tutorial reported `cc$nozero` = 0.1060861, meaning that approximately 90% of the covariance entries were zero. This value is specific to the original simulation and software configuration.

Estimation of a skew Gaussian random field

Given a realization $(y_1, \ldots, y_N)$ of the RF Y (here $y_i = y(\boldsymbol{s}_i)$), let $f_{Y_{ij}}(y_i, y_j)$ be the density of the bivariate random vector $(Y(\boldsymbol{s}_i), Y(\boldsymbol{s}_j))^T$ given by Alegría et al. (2017):

$$f_{Y_{ij}}(y_i, y_j) = 2 \sum_{\ell=1}^{2} \phi_2(\boldsymbol{y}_{ij} - \boldsymbol{\mu}_{ij}; A_\ell) \Phi_2(\boldsymbol{c}_\ell; \boldsymbol{0}, B_\ell), \tag{4}$$

where

$$\begin{aligned} A_\ell &= \sigma^2 R + \eta^2 R_\ell, \\ \boldsymbol{c}_\ell &= \eta R_\ell (\sigma^2 R + \eta^2 R_\ell)^{-1} (\boldsymbol{y}_{ij} - \boldsymbol{\mu}_{ij}), \\ B_\ell &= R_\ell - \eta^2 R_\ell (\sigma^2 R + \eta^2 R_\ell)^{-1} R_\ell, \\ \boldsymbol{\mu}_{ij} &= (\mu(\boldsymbol{s}_i), \mu(\boldsymbol{s}_j))^T, \quad \boldsymbol{y}_{ij} = (y_i, y_j)^T. \end{aligned}$$

Here R is the 2×2 latent Gaussian correlation matrix and R_1, R_2 depend on R as described by Alegría et al. (2017). Then the pairwise log-likelihood function is defined as

$$pl(\boldsymbol{\theta}) = \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \log\{f_{Y_{ij}}(y_i, y_j)\} \, w_{ij}, \tag{5}$$

where w_{ij} are non-negative weights that do not depend on $\boldsymbol{\theta}$. An efficient choice from computational and statistical viewpoints is based on neighborhoods:

$$w_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $N_k(\boldsymbol{s}_\ell)$ is the set of the k nearest neighbors of $\boldsymbol{s}_\ell$. The parameter vector is $\boldsymbol{\theta} = (\beta_1, \beta_2, \sigma^2, \eta, \alpha, \delta, \kappa)^T$. The pairwise likelihood estimator $\widehat{\boldsymbol{\theta}}_{pl}$ maximizes (5) with respect to the parameters to be estimated.

In the `GeoModels` package we can choose fixed parameters and parameters to be estimated. Pairwise likelihood estimation is performed using `GeoFit2`:

```
## estimation with pairwise likelihood
start=list(sill=sill,mean=mean,mean1=mean1,
          scale=scale,skew=skew)
fixed=list(power2=power2,nugget=nugget,smooth=smooth)
fit=GeoFit2(data=data,coordx=coords,
```

```
        corrmodel=corrmodel,X=X,neighb=3,model=model,
        start=start,fixed=fixed)
fit
```

The option `neighb=3` specifies the neighborhood order of the weight function (6), that is, $k = 3$. The function `GeoFit2` first performs a preliminary marginal initialization before the final pairwise fit. The following is an example of the output reported in the original tutorial (estimated values may vary across package versions):

```
Maximum Composite-Likelihood Fitting of Skew Gaussian Random Fields
Setting: Marginal Composite-Likelihood
Model: SkewGaussian
Type of the likelihood objects: Pairwise
Covariance model: GenWend
Optimizer: Nelder-Mead
Number of spatial coordinates: 1200
Number of estimated parameters: 5
Type of convergence: Successful
Maximum log-Composite-Likelihood value: -14542.22
Estimated parameters:
   mean mean1 scale sill skew
 0.7149 -0.5554 0.2083 1.6307 -3.3560
```

Checking model assumptions

Given the estimated regression location, variance and skewness parameters, the *standardized residuals* should be defined using the true marginal mean and variance. Because $\mu(\boldsymbol{s}) = X(\boldsymbol{s})^T\boldsymbol{\beta}$ is the skew-Gaussian *location parameter*, the marginal mean is $m_Y(\boldsymbol{s}) = X(\boldsymbol{s})^T\boldsymbol{\beta} + \eta\sqrt{2/\pi}$. Hence

$$\widehat{\epsilon}(\boldsymbol{s}_i) = \frac{y(\boldsymbol{s}_i) - X(\boldsymbol{s}_i)^T\widehat{\boldsymbol{\beta}} - \widehat{\eta}\sqrt{2/\pi}}{[\widehat{\sigma}^2 + \widehat{\eta}^2(1-2/\pi)]^{1/2}}, \quad i = 1, \ldots, N. \tag{7}$$

The resulting field has unit marginal variance and mean zero under the fitted model; its standardized marginal distribution remains skewed rather than Gaussian. Residuals can be computed using `GeoResiduals`:

```
res=GeoResiduals(fit) # computing residuals
```

Then the marginal distribution assumption on the residuals can be checked, for instance, with a QQ-plot (Figure 1, left) using `GeoQQ`:

```
### checking model assumptions: marginal distribution
GeoQQ(res)
GeoQQ(res,type="D",ylim=c(0,0.3))
```

The correlation model assumption can be checked by comparing the empirical and estimated semivariogram functions using `GeoVariogram` and `GeoCovariogram` (Figure 1, right):

```
### checking model assumptions: semivariogram model
vario=GeoVariogram(data=res$data,
                           coordx=coords,maxdist=0.5)
GeoCovariogram(res,show.vario=TRUE,
                       vario=vario,pch=20)
```

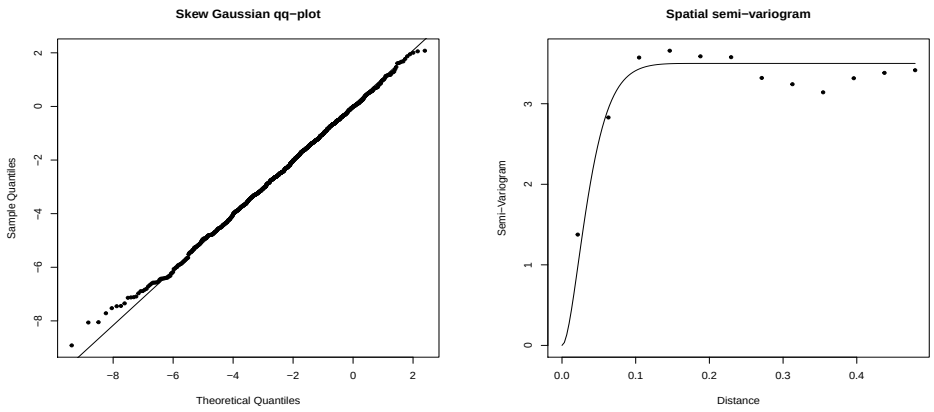


Figure 1: Left: QQ-plot for the residuals of the skew Gaussian RF. Right: empirical vs estimated semivariogram function for the residuals (figure from the original tutorial).

Prediction of skew Gaussian random fields

For a given spatial location $\boldsymbol{s}_0$ with covariates $X(\boldsymbol{s}_0)$, the optimal *linear* predictor of the skew-Gaussian RF is given by

$$\widehat{Y}(\boldsymbol{s}_0) = \widehat{m}_Y(\boldsymbol{s}_0) + \sum_{i=1}^{N} \lambda_i \big[y(\boldsymbol{s}_i) - \widehat{m}_Y(\boldsymbol{s}_i) \big], \tag{8}$$

where $\widehat{m}_Y(\boldsymbol{s}) = X(\boldsymbol{s})^T\widehat{\boldsymbol{\beta}}+\widehat{\eta}\sqrt{2/\pi}$ is the estimated marginal mean, and $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_N)^T = C^{-1}\boldsymbol{c}$, with

- $\boldsymbol{c} = (\text{cor}(Y(\boldsymbol{s}_0), Y(\boldsymbol{s}_1)), \ldots, \text{cor}(Y(\boldsymbol{s}_0), Y(\boldsymbol{s}_N)))^T$;
- $C = [\text{cor}(Y(\boldsymbol{s}_i), Y(\boldsymbol{s}_j))]_{i,j=1}^N$.

Both C and $\boldsymbol{c}$ can be computed using (2). The predictor is the best linear predictor under the fitted parameters; for a non-Gaussian field, it need not coincide with the full conditional expectation. We use `GeoKrig`, which computes linear prediction and its plug-in mean-square error.

We first set the prediction locations and associated covariates. As in the original simulation, covariates at new locations are generated independently and assumed observed there; this produces a random covariate pattern in the prediction map:

```
xx=seq(0,1,0.012)
loc_to_pred=as.matrix(expand.grid(xx,xx))
Nloc=nrow(loc_to_pred)
Xloc=cbind(rep(1,Nloc),runif(Nloc))
```

Then prediction with the fitted parameters can be performed using sparse algorithms:

```
pr=GeoKrig(fit,loc=loc_to_pred,
                    Xloc=Xloc,mse=TRUE,sparse=TRUE)
```

We compare the simulated data with predictions and associated MSE using

```
par(mfrow=c(1,3))
quilt.plot(coords,data,main="Data")
map=matrix(pr$pred,ncol=length(xx))
quilt.plot(loc_to_pred,as.vector(map),
                xlab="",ylab="",main="Simple Kriging")
map_mse=matrix(pr$mse,ncol=length(xx))
quilt.plot(loc_to_pred,as.vector(map_mse),
                xlab="",ylab="",main="mse")
```

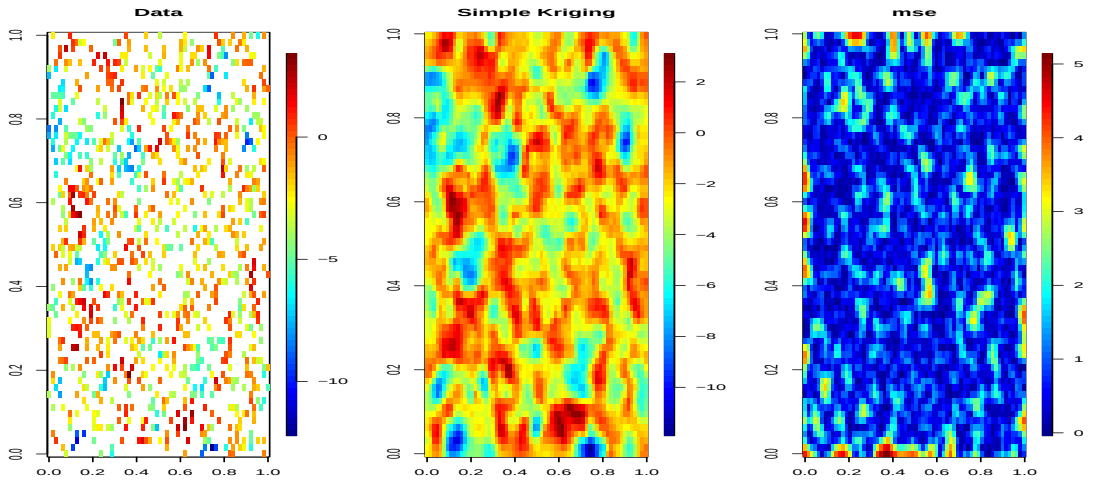


Figure 2: From left to right: observed spatial data, associated kriging map and mean-square error map for the skew Gaussian RF (figure from the original tutorial).

References

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Azzalini, A. and Capitanio, A. (2014). *The Skew-Normal and Related Families*. Cambridge University Press, Cambridge.

Bevilacqua, M., Faouzi, T., Furrer, R., and Porcu, E. (2019). Estimation and prediction using generalized Wendland covariance functions under fixed domain asymptotics. *The Annals of Statistics*, 47(2):828–856.

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